

NPG Annealings Script

Explanation and Operating Logic

File analysed: 4. NPG Annealings_NEW TRY_v3 copia.py

Prepared for the ICN2 NPG chamber automation workflow

Purpose of this PDF: This document explains what the script does, how its logic is structured, how the GUI and hardware communication work, what safety actions are active, what files are saved, and how to operate the script in practice.

Section	What it explains
1	High-level purpose and role in the NPG synthesis sequence
2	Startup configuration and hardware used by the script
3	Complete phase-by-phase annealing workflow
4	Parallel Keysight current ramp-down
5	Live GUI layout and editable controls
6	PID communication and setpoint writing
7	Data logging and saved output files
8	Abort, failsafe and shutdown behaviour
9	Operator checklist and troubleshooting
10	Function map and important implementation notes

1. Executive summary

This script runs the final NPG annealing stage after the DP-DBBA evaporation script. It controls the external oven PID setpoint through the RKC serial protocol, monitors the oven temperature, monitors the CK-1 crucible temperature from Arduino, monitors the Keysight power supply, and performs a controlled evaporator-current ramp-down in parallel with the annealing temperature sequence.

- **Main experimental role:** perform the programmed NPG annealing recipe after molecule evaporation.
- **Main controlled device:** the oven PID on COM9, whose S1 setpoint is changed through the script.
- **Parallel current action:** the Keysight current is ramped down from the starting value toward 0 A while the annealing recipe runs.
- **Operator interaction:** the GUI lets the user edit first/second stage temperatures and hold times live, and provides an Abort / Safe Stop button.
- Normal finish: PID SV is held at 0 deg C for 10 min; final plots/data are saved, and the script ends after ramp-down and evaporator switch-off confirmation.

Key idea: This is not an evaporation-thickness script. It does not read QMBs and does not stop based on thickness. Its job is the final two-stage annealing recipe plus safe current ramp-down.

2. Where this script fits in the full workflow

Step	Script / action	Role
1	Heat up + Calibration	Calibrates the CK-1 evaporation geometry and prepares working conditions.
2	Sputtering-Annealing	Handles the sputtering/annealing cycles before molecular evaporation, depending on the workflow.
3	DP-DBBA Evaporation	Evaporates the DP-DBBA precursor and finishes by leaving the Keysight at base current for the next step.
4	NPG Annealings - this script	Runs the NPG annealing recipe and performs the final controlled Keysight ramp-down.

3. Startup configuration and fixed parameters

At startup the script asks only for the run name. It then creates an output folder inside the base folder NPG Annealings.

```
RUN_NAME_PROMPT = "Enter the name for this NPG annealing run: "  
OUTPUT_BASE_FOLDER = "NPG Annealings"  
self.output_dir = os.path.join(OUTPUT_BASE_FOLDER, f"NPG Annealings {run_name}")
```

Item	Value / explanation
Initial wait	5 min at 200 deg C
First stage target	350 deg C by default, editable in the GUI
First stage hold	15 min by default, editable in the GUI
Second stage target	600 deg C by default, editable in the GUI
Second stage hold	40 min by default, editable in the GUI
Stage reached margin	1 deg C. A stage is considered reached when oven PV >= target - 1 deg C.
Cooldown target	0 deg C
Post-cooldown wait	10 min at 0 deg C before finishing
Final PID setpoint	0 deg C

4. Hardware interfaces used by the script

Device / signal	Port	Baud	What the script does
Oven PID	COM9	9600	Reads M1 process value and writes S1 setpoint using the RKC protocol.

Device / signal	Port	Baud	What the script does
Keysight power supply	COM17	9600	Configures remote mode, voltage/current protections, monitors current/voltage, ramps current down and switches output off.
Arduino CK-1 temperature	COM3	9600	Reads CK-1 crucible temperature for display and logging only.

Important limitation: The script does not control the leak valve, sputter gun, QMBs, shutter, chamber pressure, or turbo/scroll pumps. It is focused on oven annealing plus evaporator-current ramp-down.

5. Complete phase-by-phase annealing workflow

The central annealing sequence is implemented in `run_annealing_sequence()`. The phase name is stored in `SharedState` and shown in the GUI title/status panel.

Phase	PID setpoint / condition	What happens
INITIAL_WAIT	Set S1 to 200 deg C and wait 5 min	This is an active initial wait: the script writes 200 deg C to the PID, then holds for INITIAL_WAIT_S.
RAMP_TO_FIRST	Set S1 to first target, default 350 deg C	The script waits until oven PV reaches target - 1 deg C.
HOLD_FIRST	Hold first target, default 15 min	The hold duration can be edited live. The remaining time is recalculated from the live duration.
RAMP_TO_SECOND	Set S1 to second target, default 600 deg C	The script waits until oven PV reaches target - 1 deg C.
HOLD_SECOND	Hold second target, default 40 min	The hold duration can be edited live while the phase is running.
RAMP_TO_0	Set S1 to 0 deg C	The script commands the oven PID target to 0 deg C.
HOLD_0_AND_FINALIZE	Hold 0 deg C for 10 min	The script holds PID SV at 0 deg C for 10 min, then marks the annealing sequence finished.
END_CONDITION	Keep S1 at 0 deg C	No additional PID setpoint is sent before marking the run finished.
FINISHED	No new setpoint	The finished flag is set, plots and summaries are saved, and the script exits when ramp-down and switch-off confirmation are complete.

```
state.set_phase("INITIAL_WAIT", "Initial wait at 200 deg C", duration_s=INITIAL_WAIT_S)
pid.set_setpoint_c(INITIAL_WAIT_TARGET_C)
...
state.set_phase("RAMP_TO_FIRST", "Setting PID target to first-stage value")
pid.set_setpoint_c(first_target)
wait_until_temperature_reached(...)
...
state.set_phase("HOLD_0_AND_FINALIZE", "Holding PID SV at 0 deg C for 10 min")
hold_for_seconds(state, POST_COOLDOWN_WAIT_S, final_hold_message)
state.anneal_finished_normally = True
```

How a ramp phase ends: For the first and second oven ramps, the script does not wait for the exact target value. It uses target - 1 deg C as the threshold, controlled by `STAGE_REACHED_MARGIN_C = 1.0`.

6. Parallel Keysight current ramp-down

The Keysight ramp-down is started as its own thread at the same time as the oven sequence. This means the oven annealing recipe and evaporator current ramp-down occur in parallel.

Ramp-down parameter	Value	Meaning
KEYSIGHT_RAMPDOWN_STEP_A	0.005 A	Current is decreased by 5 mA per step.
KEYSIGHT_RAMPDOWN_STEP_S	15 s	Time between ramp-down steps.
FIRST_RAMPDOWN_STEP_DELAY_S	10 s	Delay before the first current decrement.
KEYSIGHT_ZERO_THRESHOLD_A	0.003 A	Current at or below this is treated as effectively zero.
KEYSIGHT_VOLTAGE_LIMIT_V	2.40 V	Voltage limit and OVP value used in the Keysight configuration.
KEYSIGHT_OCP_A	0.685 A	Overcurrent protection value configured in the Keysight.

```
def rampdown_keysight(keysight, state):
    keysight.configure_for_remote_rampdown()
    start_current = state.initial_keysight_current_a or keysight.measure_current_a()
    ...
    next_current = max(0.0, round(current_setpoint - KEYSIGHT_RAMPDOWN_STEP_A, 3))
    keysight.set_current_a(next_current)
    ...
    if current_setpoint <= 0.0:
        keysight.shutdown_output()
        state.show_switch_off_prompt()
```

Operator warning: When the current reaches zero, the script switches the Keysight output OFF and shows SWITCH OFF EVAPORATOR CURRENT. The warning is intended to disappear only after the current/voltage readings are no longer available, meaning the evaporator electronics have been switched off or disconnected.

7. Live GUI layout and controls

The GUI is a Matplotlib window with four live plots and a right-side control panel.

GUI element	What it shows or controls
Top title banner	Current phase, color-coded, with remaining time when the current phase has a known deadline.
Oven PID temperature plot	Live M1 oven process temperature read from the PID.
CK-1 temp plot	Arduino CK-1 crucible temperature. This is displayed and logged, not used to decide oven phase transitions.
Evaporator current plot	Keysight measured current during ramp-down.
Evaporator voltage plot	Keysight measured voltage during ramp-down.
First-stage target textbox/button	Changes the first oven stage target. If currently in RAMP_TO_FIRST or HOLD_FIRST, the new setpoint is sent to the PID immediately.
Second-stage target textbox/button	Changes the second oven stage target. If currently in RAMP_TO_SECOND or HOLD_SECOND, the new setpoint is sent to the PID immediately.
First/second hold time textboxes	Change the hold durations. If the relevant hold is currently running, the remaining time is recalculated.
ABORT / SAFE STOP button	Sets abort state, stops threads, sends oven PID to 0 deg C as best effort and switches Keysight output OFF.
Current status block	Phase, elapsed/remaining time, oven setpoint/PV, CK-1 temp, current, voltage, active targets, output-off state and status message.
Phase sequence block	Text summary of the planned recipe using the current editable values.
Last action block	Most recent GUI action or error message.

Live editing behavior: Editing a temperature target updates the stored recipe immediately. It also sends the new S1 setpoint immediately only if the script is currently in the corresponding ramp or hold phase.

8. Oven PID communication logic

The PIDController class uses the same RKC-style serial protocol as the other chamber scripts. It reads process value M1, reads/writes setpoint S1, and uses an XOR BCC checksum for write frames.

PID operation	Implementation
Read M1	Send EOT + address + identifier + ENQ, parse the returned STX...ETX data frame.
Read S1 before writing	Required so the script can infer the numeric formatting used by the PID.
Format target	pid_format_target_like_current_data() preserves width/decimals from the existing S1 data.
Write S1	Send EOT + address + STX + S1 + formatted data + ETX + BCC.
Verify normal write	After writing, the script rereads S1 and requires the readback to be within 0.51 deg C of the requested target.
Abort write	set_setpoint_c_best_effort() writes 0 deg C and only requires ACK, because abort should prioritize shutdown.

```
def set_setpoint_c(self, target_c):
    s1_before, sv_before = self.read_value("S1")
    data_text = pid_format_target_like_current_data(s1_before["data"], target_c)
    write_reply = self.write_s1(data_text)
    _, sv_after = self.read_value("S1")
    if sv_after is None or abs(sv_after - target_c) > 0.51:
        raise RuntimeError("PID did not confirm the requested setpoint")
```

9. Data logging and output files

All data are saved inside the run folder:

```
NPG Annealings/
  NPG Annealings <run_name>/
```

Output file	Content
<run_name>__PID_temperature_data__<date>.db	SQLite database with table temperatures(timestamp, temperature).
01__PID_temperature_data.csv	CSV containing timestamp and oven PID temperature.
<run_name>_telemetry.csv	Periodic telemetry rows with phase, oven temperature, oven setpoint, CK-1 temperature, Keysight current/voltage and message.
01__PID_temperature_curve.png	Clean standalone plot of the oven PID temperature curve, saved at shutdown.
<run_name>_final_plot.png	Screenshot-like final version of the live GUI with all four plots and status panel.
<run_name>_summary.txt	Final summary: phase, whether annealing finished normally, setpoints, latest temperatures, currents and message.

Implementation note: In this script version the telemetry CSV header lists seven columns, while log_row writes an additional CK-1 temperature value. Spreadsheet programs may therefore show one shifted/unlabelled column. The values are still written, but the header should ideally be updated.

10. Abort, failsafe and shutdown behaviour

Situation	What the script does
Normal finish	The oven sequence holds 0 deg C for 10 min; anneal_finished_normally becomes True, final files are saved and hardware connections close.
GUI Abort / Safe Stop	Marks abort, sets phase ABORTING, stops the run, then attempts to send oven PID to 0 deg C and switch Keysight output OFF.
Terminal q	Marks manual abort and stops the run.
KeyboardInterrupt / SIGINT	Marks abort and triggers the shutdown path.
Process exit before normal finish	atexit failsafe marks abort and attempts hardware shutdown.
Keysight reaches zero current	The script sends CURR 0.000 and OUTP OFF, then shows the switch-off warning.

Safety philosophy: The abort path is best-effort but conservative: move the oven PID to 0 deg C and force the Keysight output off. The script also installs a process-exit failsafe so an early Python exit does not leave the hardware unintentionally active.

11. Operator checklist

Moment	Checklist
Before running	Confirm COM9, COM17 and COM3 are correct; confirm the Keysight and PID are powered and communicating; confirm the evaporator state inherited from script 3.
Startup	Enter a clear run name. If automatic current readback fails, enter the starting Keysight current manually.
During run	Watch the phase banner, oven PV/SV, current ramp-down, CK-1 temperature and warning/status messages.
If editing recipe	Use the GUI textboxes and Apply buttons. Check Last action to confirm whether the target or hold time was accepted.
When current reaches zero	Follow the warning: switch off the evaporator/current electronics physically if required by the setup.
If something is wrong	Use ABORT / SAFE STOP rather than closing the Python window. This gives the script a chance to set PID=0 deg C and switch Keysight OFF.
After run	Check the summary file, PID temperature curve and final plot in the output folder.

12. Troubleshooting guide

Problem	Likely cause	What to check
Script fails at startup	A serial port is unavailable or wrong.	Check Windows Device Manager and update PID_PORT, KEYSIGHT_PORT or CK1_ARDUINO_PORT.
PID setpoint write fails	PID did not return S1 data or readback does not match target.	Check PID address 00, baudrate 9600, wiring, and whether the PID is already used by another program.
Cannot read Keysight current automatically	Keysight communication failed before ramp-down.	The script will ask you to enter the current manually. Also check COM17 and remote interface settings.
Switch-off warning does not disappear	The serial device may still be returning 0 current/voltage instead of disappearing.	The current logic hides the warning after missing readbacks, not simply after OUTP OFF.
Hold time edit seems not applied	The phase might not be the corresponding hold phase yet, or the textbox value was invalid.	Check Last action. Hold times are live during HOLD_FIRST/HOLD_SECOND.
Temperature target edit does not change PID immediately	The script is not currently in the corresponding ramp/hold phase.	The stored value is updated for later. Immediate PID write only occurs in RAMP_TO_FIRST/HOLD_FIRST or RAMP_TO_SECOND/HOLD_SECOND.
Final plot looks cluttered	The final plot saves the live GUI, including status panel.	Use 01_PID_temperature_curve.png for the clean report-style PID curve.

13. Function map

Function / class	Purpose
SharedState	Central state object: phase, timers, messages, editable targets, data arrays and hardware flags.
PIDController	Reads M1, reads/writes S1, verifies PID setpoint changes and handles best-effort abort setpoint.
KeysightController	SCPI communication with Keysight: configure remote mode, read current/voltage, set current and shut output down.
AnnealLogger	Writes SQLite PID temperature data, PID CSV data and periodic telemetry CSV rows.
monitor_oven	Reads oven PID M1 every READ_INTERVAL_S and logs it.
monitor_ck1_temperature	Reads CK-1 Arduino temperature for display/logging.
monitor_keysight	Reads current and voltage, detects near-zero current and shows the switch-off prompt.
rampdown_keysight	Parallel controlled current ramp-down from initial current to 0 A.
run_annealing_sequence	Main two-stage oven annealing recipe.
build_figure / update_figure	Create and refresh the live Matplotlib GUI.
save_pid_temperature_curve	Generates the clean report-style PID curve PNG at the end.
App	Top-level coordinator: creates hardware controllers, starts threads, manages GUI loop and shutdown.

14. Practical interpretation

In practical terms, the script is the final automated recipe manager for the NPG annealing step. The oven PID follows a programmable two-stage temperature sequence, while the Keysight current is progressively reduced in parallel. The GUI gives the operator enough live control to adjust the two important annealing temperatures and hold times without editing the source code, and the shutdown logic is designed to leave a traceable record of the run while reducing the risk of leaving the oven or power supply in an unsafe state.

Most important thing to remember: Use normal completion when the recipe runs correctly. Use ABORT / SAFE STOP if you need to interrupt the run, because closing the window directly gives the script less opportunity to perform the intended hardware shutdown.